

Vol 4 Chapter 6 — CMB Structure

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2026-05-23 Saturday

Chapter 6 — CMB Structure

Legacy chapter (May 2026). Last accuracy-checked 2026-09-11 (Keeper, K1892): the presentation scan found no retired reading here, and the tiers in the front matter are the May tiers, not re-verified line by line. Where this chapter and the Spine (Curriculum/Spine_DIV5_QM_GR_SM/) disagree, the Spine is current; where the Spine and the theorem registry disagree, the registry wins.

The cosmic microwave background (CMB) is the relic radiation from the early universe, measured to extraordinary precision by the COBE, WMAP, and Planck satellites. Its temperature spectrum, polarization patterns, and density-fluctuation distribution carry information about cosmic parameters that BST derives from substrate primaries.

6.1 The scalar spectral index n_s

$$n_s = 1 - \frac{n_C}{N_{\max}} = 1 - \frac{5}{137} \approx 0.9635.$$

Planck measurement: $n_s = 0.9649 \pm 0.0042$. BST prediction matches at 0.3σ . The $5/137 = n_C/N_{\max}$ structural reading: the CMB scalar spectral tilt is the substrate's $N_{\max}$ inheritance at the cosmological scale.

Elie Toy 1401 (7/8 sub-checks passing) verifies the computational match.

6.2 Matter and dark energy densities

$$\Omega_\Lambda = \frac{N_c + 2n_C}{N_c^2 + 2n_C} = \frac{13}{19} \approx 0.684,$$

$$\Omega_m = \frac{C_2}{N_c^2 + 2n_C} = \frac{6}{19} \approx 0.316.$$

Planck measurements: $\Omega_\Lambda = 0.685 \pm 0.007$, $\Omega_m = 0.315 \pm 0.007$. Both BST predictions match at 0.07σ — well within experimental precision.

The ratio $\Omega_\Lambda + \Omega_m = 19/19 = 1$ structurally closes — a flat universe forced by substrate integer combinations.

6.3 The CMB temperature

$$T_{\text{CMB}} = 2.737 \text{ K (BST prediction),}$$

against measured 2.7255 K — match at 0.43%. Elie Toy 904 verifies. The substrate-derivation runs through cosmological Bergman-vacuum scaling.

6.4 The CMB acoustic peaks

The full CMB temperature anisotropy spectrum, including the acoustic peak positions and heights, is substrate-derived via the same Λ CDM machinery that Route C of Chapter 5 ran through CAMB. Per-peak match precisions are at the percent level across the full multipole spectrum.

6.5 What comes next

Chapter 7 covers inflation parameters with substrate-derived predictions for tensor-to-scalar ratio r , tensor spectral index n_t , and running α_s .

Where to look this up: n_s derivation: Elie Toy 1401. Ω_Λ and Ω_m : Tasks #83 + #100 + #101. CMB temperature: Toy 904. For Planck CMB measurements: Planck Collaboration 2018, *Astronomy & Astrophysics* 641:A6.